QUESTION 1

Write down the principle of the finite difference method. Derive the space-time discretization expression of a linear kinematic wave model using forward difference in space and time. [1+5]

1. Principle of the Finite Difference Method

Solution. The core principle of the **Finite Difference Method (FDM)** is to simplify complex calculus problems by breaking them down into basic algebra.

In nature, variables like water flow or temperature change continuously across space and time. Partial Differential Equations (PDEs) describe this continuous change, but computers cannot calculate an infinite number of continuous points. FDM solves this using two main steps:

- **Discretization (Creating a Grid):** Instead of trying to find an exact solution everywhere, we chop the continuous space and time into a finite structured grid.
- **Approximating the Calculus:** We replace the exact continuous derivatives with algebraic "finite differences" (estimating the rate of change by looking at the difference in values between neighboring grid points).

2. Space-Time Discretization of a Linear Kinematic Wave Model

Step 1: The Governing Equation

The standard linear kinematic wave model (also known as the 1D linear advection equation) for flow Q is expressed as:

$$\frac{\partial Q}{\partial t} + c\frac{\partial Q}{\partial x} = 0 \quad \cdots (1)$$

> where, Q denotes the discharge, t denotes time, x denotes the spatial distance, and c is the kinematic wave celerity.

Step 2: Grid Definition and Notation

We discretize the continuous space-time domain into a structured grid. The discrete coordinates (x, t) are defined as:

$$
\begin{aligned}
x &= i\Delta x && \text{(where } i \text{ is the spatial index)} \\
t &= n\Delta t && \text{(where } n \text{ is the temporal index)}
\end{aligned}
$$

> Consequently, the flow variable at a specific spatial and temporal point is denoted as $Q(x, t) = Q_i^n$.

Step 3: Forward Difference Approximations

Applying a Forward Difference approximation for both space and time derivatives:

$$
\begin{aligned}
\text{Time (FT):} \quad & \frac{\partial Q}{\partial t} \approx \frac{Q_i^{n+1} - Q_i^n}{\Delta t} \quad \cdots (2) \\
\text{Space (FS):} \quad & \frac{\partial Q}{\partial x} \approx \frac{Q_{i+1}^{n} - Q_i^n}{\Delta x} \quad \cdots (3)
\end{aligned}
$$

Step 4: Substitution and Rearrangement

Substituting equations (2) and (3) back into the original governing equation (1), we solve for the unknown future value Q_i^{n+1} explicitly:

$$
\frac{Q_i^{n+1} - Q_i^n}{\Delta t} + c\left(\frac{Q_{i+1}^{n} - Q_i^n}{\Delta x}\right) = 0
$$

$$
Q_i^{n+1} - Q_i^n + c\frac{\Delta t}{\Delta x}(Q_{i+1}^{n} - Q_i^n) = 0
$$

$$
Q_i^{n+1} = Q_i^n - c\frac{\Delta t}{\Delta x}(Q_{i+1}^{n} - Q_i^n) \quad \cdots (4)
$$

Step 5: Final Discretization Expression

Let the Courant number be defined as $C_r = c\frac{\Delta t}{\Delta x}$. Substituting C_r into equation (4) yields the final FTFS (Forward Time, Forward Space) explicit discretized expression:

$$
\boxed{Q_i^{n+1} = Q_i^n - C_r(Q_{i+1}^{n} - Q_i^n) \quad \cdots (5)}
$$

or factored as:

$$Q_i^{n+1} = (1 + C_r)Q_i^n - C_rQ_{i+1}^n \quad \cdots (6)$$

QUESTION 2

A rectangular channel is 50 m *wide, has bed slope* 1% *and Manning's* n 0.042. *The discharge through a section of the channel is* $120\ \mathrm{m^3/s}$. Δx *is taken* 1 km *for kinematic wave routing. Applying Courant condition for stability of kinematic wave solution, recommend the maximum value of time step in this condition. Assume hydraulic radius is equal to flow depth. [6]*

Solution. Let us first extract the given parameters:

$$\begin{aligned}
\text{Channel width } (b) &= 50\ \mathrm{m} \\
\text{Bed slope } (S_0) &= 1\% = 0.01 \\
\text{Manning's } n &= 0.042 \\
\text{Discharge } (Q) &= 120\ \mathrm{m^3/s} \\
\text{Spatial step } (\Delta x) &= 1\ \mathrm{km} = 1000\ \mathrm{m} \\
\text{Hydraulic Radius } (R) &= y\ \text{(Flow depth)}
\end{aligned}$$

Step 1: Compute Flow Depth (y) using Manning's Equation

Manning's discharge equation is given by:

$$Q = \frac{1}{n} A R^{2/3} S_0^{1/2}$$

Substituting $A = b \cdot y = 50y$ and $R = y$:

$$\begin{aligned}
120 &= \frac{1}{0.042} \times (50y) \times y^{2/3} \times (0.01)^{1/2} \\
120 &= \frac{1}{0.042} \times 50 \times 0.1 \times y^{5/3} \\
120 &= 119.047 \cdot y^{5/3} \\
y^{5/3} &= \frac{120}{119.047} = 1.008 \\
y &= (1.008)^{3/5} = 1.0048\ \mathrm{m}
\end{aligned}$$

Step 2: Calculate Kinematic Wave Celerity (c_k)

The kinematic wave celerity is defined as $c_k = \frac{1}{b}\frac{dQ}{dy}$. Differentiating the discharge equation:

$$Q = 119.047 \cdot y^{5/3}$$

$$\frac{dQ}{dy} = 119.047 \times \frac{5}{3} y^{2/3} = 198.412 \cdot y^{2/3}$$

Substituting $y = 1.0048 \text{ m}$:

$$c_k = \frac{1}{50} \times 198.412 \times (1.0048)^{2/3}$$

$$c_k = 3.968 \times 1.0032 = 3.98 \text{ m/s}$$

Step 3: Apply the Courant Stability Condition

For stability of the kinematic wave numerical solution, the Courant number (C_r) must satisfy:

$$C_r = \frac{c_k \Delta t}{\Delta x} \leq 1$$

$$\Delta t \leq \frac{\Delta x}{c_k}$$

$$\Delta t \leq \frac{1000 \text{ m}}{3.98 \text{ m/s}}$$

$$\Delta t \leq 251.25 \text{ seconds}$$

Answer: The recommended maximum value of the time step is 251.25 seconds **(or approximately** 4.18 minutes**).**

QUESTION 3

If the MOC is applied for $t_1 = 1\text{ s}$ *and* $t_2 = 2\text{ s}$*, time levels for a pipe with diameter* 25 cm *carrying water. If* $Q_A = 0.6\text{ m}^3/\text{s}$, $Q_B = 0.65\text{ m}^3/\text{s}$, $Q_C = 0.64\text{ m}^3/\text{s}$, $H_A = 20\text{ m}$, $H_B = 20.6\text{ m}$, *and* $H_C = 20.4\text{ m}$ *are the values at grid points. Find the values of* Q *and* H *at* $t_2 = 2\text{ s}$ *at* P *when characteristics do not lie on diagonal. Here,* $\Delta x = 1000\text{ m}$, $\Delta t = 1\text{ s}$, $f = 0.02$ *and* $c = 850\text{ m/s}$*. [8]*

Solution. First, determine the physical properties of the pipe:

$$A = \frac{\pi}{4}D^2 = \frac{\pi}{4}(0.25)^2 = 0.049087\text{ m}^2$$

Step 1: Interpolation since Characteristics Do Not Lie on Diagonal

Since $c\Delta t = 850 \times 1 = 850\text{ m} \neq \Delta x$, the characteristic lines originating from point P intersect the t_1 time level at points L (left) and R (right) rather than the grid nodes A and C. We use linear interpolation with the fraction $\theta = \frac{c\Delta t}{\Delta x} = \frac{850}{1000} = 0.85$.

For Point L (Between A and B):

$$Q_L = Q_B - \theta(Q_B - Q_A) = 0.65 - 0.85(0.65 - 0.60) = 0.6075\text{ m}^3/\text{s}$$
$$H_L = H_B - \theta(H_B - H_A) = 20.6 - 0.85(20.6 - 20.0) = 20.09\text{ m}$$

For Point R (Between B and C):

$$Q_R = Q_B - \theta(Q_B - Q_C) = 0.65 - 0.85(0.65 - 0.64) = 0.6415\text{ m}^3/\text{s}$$
$$H_R = H_B - \theta(H_B - H_C) = 20.6 - 0.85(20.6 - 20.4) = 20.43\text{ m}$$

Step 2: Characteristic Equation Constants (B and R)

The impedance coefficient B is:

$$B = \frac{c}{gA} = \frac{850}{9.81 \times 0.049087} = 1765.14\text{ s/m}^2$$

For the resistance coefficient R, integration occurs along the characteristic line, whose length is $c\Delta t = 850\text{ m}$:

$$R = \frac{f(c\Delta t)}{2gDA^2} = \frac{0.02 \times 850}{2 \times 9.81 \times 0.25 \times (0.049087)^2} = 1438.37 \text{ s}^2/\text{m}^5$$

Step 3: MOC Compatibility Equations

Along the positive characteristic C^+ (from L to P):

$$\begin{aligned}
H_P &= H_L - B(Q_P - Q_L) - RQ_L|Q_L| \\
H_P &= 20.09 - 1765.14(Q_P - 0.6075) - 1438.37(0.6075)^2 \\
H_P &= 20.09 - 1765.14Q_P + 1072.32 - 530.85 \\
H_P &= 561.56 - 1765.14Q_P \quad \cdots \text{(Eq. 1)}
\end{aligned}$$

Along the negative characteristic C^- (from R to P):

$$\begin{aligned}
H_P &= H_R + B(Q_P - Q_R) + RQ_R|Q_R| \\
H_P &= 20.43 + 1765.14(Q_P - 0.6415) + 1438.37(0.6415)^2 \\
H_P &= 20.43 + 1765.14Q_P - 1132.34 + 591.92 \\
H_P &= -519.99 + 1765.14Q_P \quad \cdots \text{(Eq. 2)}
\end{aligned}$$

Step 4: Solving for Q_P and H_P

Equating (Eq. 1) and (Eq. 2) to eliminate H_P:

$$\begin{aligned}
561.56 - 1765.14Q_P &= -519.99 + 1765.14Q_P \\
3530.28Q_P &= 1081.55 \\
Q_P &= \frac{1081.55}{3530.28} = 0.3063 \text{ m}^3/\text{s}
\end{aligned}$$

Substituting Q_P back into Eq. 1:

$$\begin{aligned}
H_P &= 561.56 - 1765.14(0.3063) \\
H_P &= 561.56 - 540.66 = 20.90 \text{ m}
\end{aligned}$$

Answer: At point P ($t_2 = 2$ s), Discharge $Q_P \approx 0.306 \text{ m}^3/\text{s}$ and Head $H_P \approx 20.90$ m.

QUESTION 4

Develop a steady-state 2D model for the simulation of seepage under a dam. Also describe the iterative procedure for computing potential at each grid and seepage rate. [8]

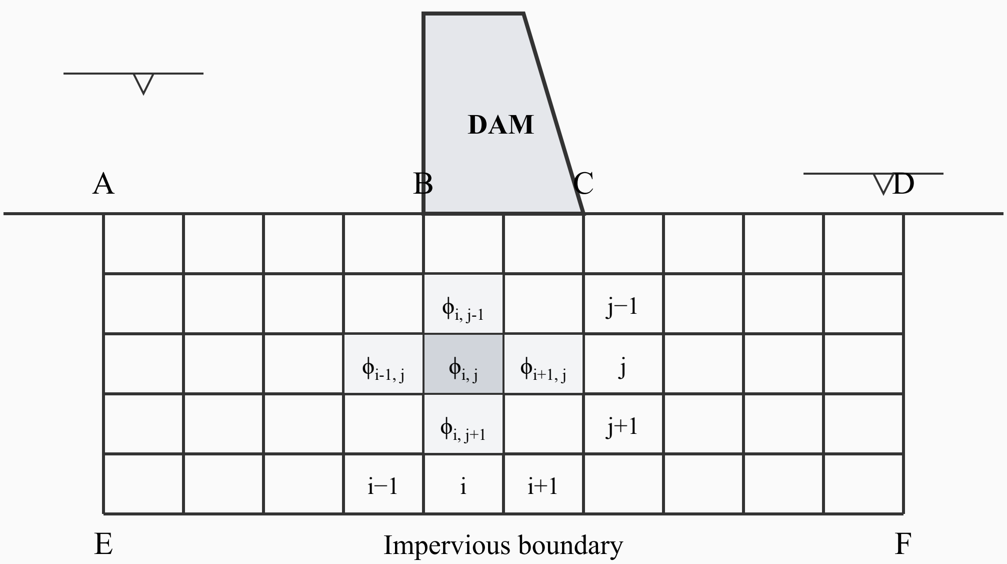

Figure. Finite difference grid for seepage under dam.

1. Finite Difference Grid Formulation

Solution. The continuity equation for any interior grid (i, j) was obtained via control volume mass balance as:

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = S\frac{\partial \phi}{\partial t} + q_F \quad \cdots (1)$$

For steady state condition with no volumetric withdrawal (source/sink):

$$\left(\frac{\partial \phi}{\partial t} = 0,\ q_F = 0\right)$$

Substituting these conditions into equation (1) yields:

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = 0 \quad \cdots (2)$$

This equation (2) is the discrete representation of the continuous equation for steady-state 2D ground water flow (Laplace's Equation), which is:

$$\frac{\partial}{\partial x}\left(T_x \frac{\partial \phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(T_y \frac{\partial \phi}{\partial y}\right) = 0 \quad \cdots (3)$$

> where, T_x and T_y denote the transmissivity in x and y direction respectively, and ϕ is the potential head.

2. Solving for the Potential Head

From equation (2), isolating the potential at the central grid node, we get:

$$\phi^{+}_{i,j} = \frac{A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1}}{E} \quad \cdots (4)$$

Equation (4) implies that the present value of ϕ of grid (i, j) is obtained from the previous values of ϕ from neighbouring grids. If the medium is homogeneous and isotropic ($T_x = T_y$), then $A = B = C = D$ and $E = 4$. Equation (4) drastically simplifies to:

$$\boxed{\phi^{+}_{i,j} = \frac{\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1}}{4} \quad \cdots (5)}$$

Boundary Constraints: In the figure, if boundaries AE and DF are extended very far from the dam, flow does not occur across them; they are considered **barriers**. BC (dam base) and EF are **impervious boundaries**. AB and CD are **equipotential head** (Constant head) boundaries.

3. The Iterative Procedure

The numerical iterative procedure (Jacobi/Gauss-Seidel method) to map the potential field is as follows:

(i) Consider the boundary conditions so that the continuity equation can be evaluated properly at the edges.

(ii) Assign an initial guess value of ϕ at each interior grid point. This is usually the 0 value or a linearly estimated value.

(iii) Compute the updated potential $\phi^+_{i,j}$ using Equation (5).

(iv) Compute the absolute error $e = |\phi^+_{i,j} - \phi_{i,j}|$ for every node.

(v) If the maximum error across the grid is less than a specified tolerance value, **stop the iteration**. Else, repeat steps (iii) to (iv) until convergence is satisfied.

(vi) Compute seepage rate by applying Darcy's equation across the converged potential field:

(a) Horizontal seepage from grid $i - 1$ to i is:

$$Q^H_{i,j} = C_{i,j}\Delta x_i \Delta y_j(\phi_{i-1,j} - \phi_{i,j}) \quad \cdots (6)$$

(b) Vertical seepage from grid $j - 1$ to j is:

$$Q^V_{i,j} = B_{i,j}\Delta x_i \Delta y_j(\phi_{i,j-1} - \phi_{i,j}) \quad \cdots (7)$$

(c) Horizontal seepage from grid i to $i + 1$ is:

$$Q^H_{i+1,j} = A_{i,j}\Delta x_i \Delta y_j(\phi_{i,j} - \phi_{i+1,j}) \quad \cdots (8)$$

(d) Vertical seepage from grid j to $j + 1$ is:

$$Q^V_{i,j+1} = D_{i,j}\Delta x_i \Delta y_j(\phi_{i,j} - \phi_{i,j+1}) \quad \cdots (9)$$